

Forced Magnetic Reconnection in Tokamak Plasmas

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I. INTRODUCTION

Consider an intrinsically tearing-stable tokamak plasma that is subject to an externally generated, static (in the laboratory frame), non-axisymmetric, magnetic perturbation. The perturbation inevitably resonates at various rational surfaces inside the plasma, and acts to drive magnetic reconnection at these surfaces. This process, which is known as *forced magnetic reconnection*, generates magnetic island chains within the plasma.^{1,2} Such chains degrade the plasma energy confinement,³ and, in extreme cases, can even trigger a major disruption.

This paper investigates the drift-magnetohydrodynamical (MHD) response of an idealized cylindrical tokamak plasma to a static resonant magnetic perturbation (RMP), resonant at a single rational surface, that is suddenly switched on. The problem under discussion is an extension of one considered in a well-known paper by Hahm & Kulsrud,⁴ and a host of subsequent papers.^{5–12} The present study takes plasma flows, including diamagnetic flows, into account. Such flows are often omitted from the analysis, but can profoundly modify forced magnetic reconnection. The ion sound-radius, and realistic levels of perpendicular energy and momentum transport, are also included in the analysis.

II. FUNDAMENTAL ANALYSIS

A. Plasma Equilibrium

Our tokamak equilibrium is approximated as a periodic cylinder of circular cross-section.¹³ Let us employ a conventional set of right-handed cylindrical coordinates, r , θ , z . The equilibrium magnetic flux-surfaces lie on surfaces of constant r . The system is assumed to

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be periodic in the z (“toroidal”) direction, with periodicity length $2\pi R_0$, where R_0 is the simulated major radius of the plasma. It is helpful to define the simulated toroidal angle $\varphi = z/R_0$. Let a be the minor radius of the plasma. It is assumed that $a/R_0 \ll 1$. It is also assumed that the toroidal plasma beta parameter is $\mathcal{O}(a^2/R_0^2)$. The equilibrium magnetic field is written $\mathbf{B} = (0, B_\theta(r), B_\varphi)$, where $B_\theta(r)$ is the poloidal magnetic field-strength, and B_φ the (approximately) spatially uniform “toroidal” magnetic field-strength. The equilibrium current density takes the form $\mathbf{j} = (0, 0, j_\varphi(r))$. The *safety-factor* profile is defined as $q(r) = r B_\varphi/[R_0 B_\theta(r)]$, and is assumed to be of order unity.

B. Plasma Response to RMP

Suppose that the plasma equilibrium is subject to a relatively small amplitude (compared to the equilibrium magnetic field) RMP, with $m > 0$ periods in the poloidal direction, and $n > 0$ periods in the toroidal direction, that is switched on suddenly at $t = 0$. The perturbation resonates with the plasma at the magnetic flux-surface at which $\mathbf{k} \cdot \mathbf{B} = 0$, where $\mathbf{k} = (0, m/r, -n/R_0)$ is the wavenumber of the perturbation. It follows that the safety-factor takes the value m/n at this so-called *rational* surface.

As far as its response to the magnetic perturbation is concerned, a high Lundquist number tokamak plasma can be divided into two regions.^{1,2,4–6,9,11,12,14–16} The so-called *inner region* lies in the immediate vicinity of the rational surface. The so-called *outer region* comprises everywhere in the plasma apart from the inner region, as well as the vacuum region surrounding the plasma.

The response of the plasma to the RMP in the outer region is governed by the equations of linear, marginally-stable, ideal-MHD.¹ The minimal model that can realistically describe the response of the plasma in the inner region should contain plasma resistivity, $\mathbf{E} \times \mathbf{B}$ flows, electron and ion diamagnetic flows, the ion sound-radius, and anomalous cross-flux-surface transport of energy and momentum. It is important to include diamagnetic flows in the model because they are likely to be similar in magnitude to $\mathbf{E} \times \mathbf{B}$ flows in next-generation tokamaks. It is important to include the ion sound-radius in the model because the response of the ion fluid to the RMP is radically different to that of the electron fluid on length-scales below the ion sound-radius. Finally, it is important to include realistic levels of perpendicular energy and momentum transport in the model because these effects significantly change the

response of the inner region to the RMP.¹⁵ The solutions in the inner and outer regions must be asymptotically matched to one another.¹⁴

C. Driven Magnetic Reconnection

In this paper, all helical magnetic fluxes are normalized with respect to $r_s B_\varphi$, where r_s is the minor radius of the rational surface. Let $\Psi_b(t)$ be the helical magnetic flux associated with the RMP to which the plasma is subject. (See Sect. A 3.) Likewise, let $\Psi_0(t)$ be the driven reconnected helical magnetic flux at the rational surface. (See Sect. B 11.) Making use of the analysis of Appendices A and B, the relationship between Ψ_b and Ψ_0 is found to be

$$\Psi_0(\hat{t}) = \frac{1}{2\pi i} \int_C \frac{E_{sb} F_s(\hat{g}) \bar{\Psi}_b(\hat{g}) e^{\hat{g}\hat{t}}}{S^{1/3} \hat{\Delta}_s(\hat{g}) - E_{ss}} d\hat{g}, \quad (1)$$

where

$$\bar{\Psi}_b(\hat{g}) = \int_0^\infty \Psi_b(\hat{t}) e^{-\hat{g}\hat{t}} d\hat{t}. \quad (2)$$

Here, C is the Bromwich contour, and

$$\hat{t} = \frac{t}{S^{1/3} \tau_H}, \quad (3)$$

where S and τ_H are the Lundquist number and hydromagnetic time, respectively, at the rational surface. The real dimensionless quantities E_{ss} and E_{sb} are the tearing stability index, and the coupling constant between the RMP and the rational surface, respectively. (See Sects. A 4 and A 5.) It is assumed that $E_{ss} < 0$, which implies that the plasma is intrinsically tearing-stable, and that any reconnected magnetic flux at the rational surface is due to the action of the RMP. The complex dimensionless quantities $F_s(\hat{g})$ and $\hat{\Delta}_s(\hat{g})$ are determined by the layer solution in the inner region.

D. Linear Resonant Plasma Response Model

Our adopted linear resonant plasma response model is the three-field model presented in Ref. 16. (See Appendix B.) This model is defined by the following six dimensionless parameters: $Q_E = -S^{1/3} n \omega_E \tau_H$, $Q_e = -S^{1/3} n \omega_{*e} \tau_H$, $Q_i = -S^{1/3} n \omega_{*i} \tau_H$, $D = S^{1/3} \iota_e^{1/2} \hat{d}_\beta$, $P_\perp = \tau_R/\tau_\perp$, and $P_\varphi = \tau_R/\tau_\varphi$. Here, τ_R is the resistive diffusion timescale, τ_H the hydrodynamic timescale, $\tau_\perp$ the energy confinement timescale, τ_φ the momentum confinement

timescale, $S = \tau_R/\tau_H$ the Lundquist number, ω_E the $\mathbf{E} \times \mathbf{B}$ frequency, ω_{*e} the electron diamagnetic frequency, ω_{*i} the ion diamagnetic frequency, $d_\beta = c_\beta d_i$ the ion sound-radius, d_i the collisionless ion skin-depth, $c_\beta = \sqrt{\beta/(1+\beta)}$, $\sqrt{\beta} = \sqrt{(5/3)\mu_0 p/B_\varphi}$ the ratio of the sound speed to the Alfvén speed, and p the plasma pressure. Moreover, $P_\perp$ and P_φ are magnetic Prandtl numbers, $\iota_e = -\omega_{*e}/(\omega_{*i} - \omega_{*e})$, and $\hat{d}_\beta = d_\beta/r_s$. All quantities are evaluated at the rational surface.

E. Linear Resonant Plasma Response Regimes

According to the analysis of Appendix C, there are seven different linear resonant plasma response regimes.

The *resistive-inertial* regime is such that

$$\hat{\Delta}_s(\hat{g}) = \frac{2\pi \Gamma(3/4)}{\Gamma(1/4)} [\hat{g} + i(Q_E + Q_e)]^{3/4} (\hat{g} + iQ_E)^{1/4} [\hat{g} + i(Q_E + Q_i)]^{1/4}, \quad (4)$$

$$F_s(\hat{g}) = 1, \quad (5)$$

$$\hat{\delta} = \frac{|\hat{g} + iQ_E|^{1/4} |\hat{g} + i(Q_E + Q_i)|^{1/4}}{|\hat{g} + i(Q_E + Q_e)|^{1/4}}. \quad (6)$$

Here, the resonant layer width is $\delta = r_s S^{-1/3} \hat{\delta}$.

The *viscous-resistive* regime is such that

$$\hat{\Delta}_s(\hat{g}) = \frac{6^{2/3} \pi \Gamma(5/6)}{\Gamma(1/6)} [\hat{g} + i(Q_E + Q_e)] P_\varphi^{1/6}, \quad (7)$$

$$F_s(\hat{g}) = 1, \quad (8)$$

$$\hat{\delta} = P_\varphi^{1/6}. \quad (9)$$

The *semi-collisional* regime is such that

$$\hat{\Delta}_s(\hat{g}) = \frac{\pi [\hat{g} + i(Q_E + Q_e)] (\hat{g} + iQ_E)^{1/2}}{D}, \quad (10)$$

$$F_s(\hat{g}) = 1, \quad (11)$$

$$\hat{\delta} = \frac{|\hat{g} + iQ_E|^{1/2}}{D}. \quad (12)$$

The *diffusive-resistive* regime is such that

$$\hat{\Delta}_s(\hat{g}) = \frac{2\pi \Gamma(3/4)}{\Gamma(1/4)} \frac{[\hat{g} + i(Q_E + Q_e)] \iota_e^{1/4} P_\perp^{1/4}}{D^{1/2}}, \quad (13)$$

$$F_s(\hat{g}) = 1, \quad (14)$$

$$\hat{\delta} = \frac{\iota_e^{1/4} P_\perp^{1/4}}{D^{1/2}}. \quad (15)$$

The *inertial* regime is such that

$$\hat{\Delta}_s(\hat{g}) = -\frac{\pi}{(\hat{g} + i Q_E)^{1/2} [\hat{g} + i (Q_E + Q_i)]^{1/2}}, \quad (16)$$

$$F_s(\hat{g}) = -\frac{2}{[\hat{g} + i (Q_E + Q_e)] (\hat{g} + i Q_E) [\hat{g} + i (Q_E + Q_i)]}, \quad (17)$$

$$\hat{\delta} = |\hat{g} + i Q_E|^{1/2} |\hat{g} + i (Q_E + Q_i)|^{1/2}. \quad (18)$$

The *viscous-inertial* regime is such that

$$\hat{\Delta}_s(\hat{g}) = -\frac{\pi}{2} \frac{\Gamma(1/4)}{\Gamma(3/4)} \frac{1}{[\hat{g} - i (Q_E + Q_e)]^{1/4} P_\varphi^{1/4}}, \quad (19)$$

$$F_s(\hat{g}) = -\frac{\pi^{1/2}}{2} \frac{\Gamma(1/4)}{\Gamma(3/4)} \frac{1}{[\hat{g} - i (Q_E + Q_e)]^{3/2} P_\varphi^{1/2}}, \quad (20)$$

$$\hat{\delta} = |\hat{g} - i (Q_E + Q_e)|^{1/4} P_\varphi^{1/4}. \quad (21)$$

Finally, the *diffusive-inertial* regime is such that

$$\hat{\Delta}_s = -\frac{\pi D}{[\hat{g} + i (Q_E + Q_e)]^{1/2} \iota_e^{1/2} P_\perp^{1/2}}, \quad (22)$$

$$F_s(\hat{g}) = -\frac{2 D^2}{[\hat{g} + i (Q_E + Q_e)]^2 \iota_e P_\perp}, \quad (23)$$

$$\hat{\delta} = |\hat{g} + i (Q_E + Q_e)|^{1/2} \iota_e^{1/2} P_\perp^{1/2}. \quad (24)$$

Suppose that $|Q_E| \sim |Q_e| \sim |Q_i| \sim Q$ and $P_\varphi \sim P_\perp \sim P$. Let $R = \sqrt{\hat{g}^2 + Q^2}$. The extents of the various linear resonant response regimes in R - P space is illustrated in Figs. 1 and 2.

III. DRIVEN MAGNETIC RECONNECTION REGIMES

A. Model RMP

Suppose that the RMP takes the form

$$\Psi_b(t) = \Psi_{b0} \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}. \quad (25)$$

In other words, the RMP is switched on suddenly at $t = 0$. It follows from Eq. (2) that

$$\bar{\Psi}_b(\hat{g}) = \frac{\Psi_{b0}}{\hat{g}}. \quad (26)$$

Let

$$\Psi_0(\hat{t}) = \Psi_\infty \hat{\Psi}_0(\hat{t}), \quad (27)$$

where

$$\Psi_\infty = \frac{E_{sb} \Psi_{b0}}{(-E_{ss})} \quad (28)$$

is the reconnected magnetic flux that would be driven at the rational surface in the absence of a current sheet at the surface. In other words, Ψ_∞ is the maximum possible time-asymptotic reconnected flux. It follows from Eq. (1) and the previous three equations that

$$\hat{\Psi}_0(\hat{t}) = \frac{1}{2\pi i} \int_C \frac{F_s(\hat{g}) e^{\hat{g}\hat{t}}}{\hat{g} [\Sigma \hat{\Delta}_s(\hat{g}) + 1]} d\hat{g}, \quad (29)$$

where $\Sigma = S^{1/3}/(-E_{ss})$.

In the various non-constant- ψ resonant response regimes (i.e., the inertial, viscous-inertial, and diffusive-inertial regimes), we shall assume that $\Sigma |\Delta_s(g)| \gg 1$, so

$$\hat{\Psi}_0(\hat{t}) \simeq \frac{1}{2\pi i \Sigma} \int_C \frac{F_s(\hat{g}) e^{\hat{g}\hat{t}}}{\hat{g} \hat{\Delta}_s(\hat{g})} d\hat{g} \quad (30)$$

In the various constant- ψ regimes (i.e., the resistive-inertial, viscous-resistive, semi-collisional, and diffusive-resistive regimes), $F_s \simeq 1$, so

$$\hat{\Psi}_0(\hat{t}) \simeq \frac{1}{2\pi i} \int_C \frac{e^{\hat{g}\hat{t}}}{\hat{g} [\Sigma \hat{\Delta}_s(\hat{g}) + 1]} d\hat{g}, \quad (31)$$

B. Inertial Regime

In the inertial resonant response regime, Eqs. (16), (17), and (30) yield

$$\hat{\Psi}_{0I}(\hat{t}) = \frac{1}{2\pi i} \int_C \bar{\Psi}_{0I}(\hat{g}) e^{\hat{g}\hat{t}} d\hat{g}, \quad (32)$$

where

$$\bar{\Psi}_{0I}(\hat{g}) = \frac{\epsilon_I}{i(Q_E + Q_e)} \left[\frac{1}{\hat{g}} - \frac{1}{\hat{g} + i(Q_E + Q_e)} \right] \frac{1}{(\hat{g} + iQ_E)^{1/2} [\hat{g} + i(Q_E + Q_i)]^{1/2}}, \quad (33)$$

and

$$\epsilon_I = \frac{2}{\pi \Sigma}. \quad (34)$$

Now, if

$$\bar{\Psi}_{0I}(\hat{g}) = \frac{\bar{H}(\hat{g})}{\hat{g} + a} \quad (35)$$

then Eq. (4.1.20) of Ref. 23 yields

$$\hat{\Psi}_{0I}(\hat{t}) = \int_0^{\hat{t}} e^{a(\hat{t}' - \hat{t})} H(\hat{t}') d\hat{t}', \quad (36)$$

where $\bar{H}(\hat{g})$ is the Laplace transform of $H(\hat{t})$. In the present case,

$$\bar{H}(\hat{g}) = \frac{1}{(\hat{g} + i Q_E)^{1/2} [g + i (Q_E + Q_i)]^{1/2}} = \frac{1}{(\hat{g}^2 + \alpha \hat{g} + \beta)^{1/2}}, \quad (37)$$

where

$$\alpha = i(2Q_E + Q_i), \quad (38)$$

$$\beta = -Q_E(Q_E + Q_i). \quad (39)$$

Note that

$$\beta - \frac{1}{4} \alpha^2 = -Q_E(Q_E + Q_i) + \frac{1}{4} (2Q_E + Q_i)^2 = \frac{Q_i^2}{4}. \quad (40)$$

However Eq. (5.3.34) of Ref. 23 yields,

$$H(\hat{t}) = e^{-\alpha \hat{t}/2} J_0[(\beta - \alpha^2/4)^{1/2} \hat{t}] = e^{-i(Q_E + Q_i/2)\hat{t}} J_0\left(\frac{Q_i}{2} \hat{t}\right), \quad (41)$$

where $J_0(z)$ is a Bessel function. Thus,

$$\hat{\Psi}_{0I}(\hat{t}) = \frac{\epsilon_I}{i(Q_E + Q_e)} \int_0^{\hat{t}} \left[1 - e^{i(Q_E + Q_e)(\hat{t}' - \hat{t})}\right] e^{-i(Q_E + Q_i/2)\hat{t}'} J_0\left(\frac{Q_i}{2} \hat{t}'\right) d\hat{t}'. \quad (42)$$

At small $\hat{t}$,

$$\hat{\Psi}_{0I}(\hat{t}) \simeq \frac{\epsilon_I \hat{t}^2}{\Gamma(3)}. \quad (43)$$

At large $\hat{t}$,

$$\hat{\Psi}_{0I}(\hat{t}) \simeq \frac{\epsilon_I}{i(Q_E + Q_e)} \int_0^{\infty} \left[1 - e^{i(Q_E + Q_e)(\hat{t}' - \hat{t})}\right] e^{-i(Q_E + Q_i/2)\hat{t}'} J_0\left(\frac{Q_i}{2} \hat{t}'\right) d\hat{t}'. \quad (44)$$

But, Eq. (6.621.4) of Ref. 29 yields

$$\int_0^{\infty} e^{-\alpha x} J_0(\beta x) dx = \frac{1}{\sqrt{\alpha^2 + \beta^2}}, \quad (45)$$

so

$$\hat{\Psi}_{0I}(\hat{t}) \simeq \frac{\epsilon_I}{i(Q_E + Q_i)} \left\{ \frac{1}{(iQ_E)^{1/2} [i(Q_E + Q_i)]^{1/2}} - \frac{e^{-i(Q_E + Q_e)\hat{t}}}{(iQ_e)^{1/2} [i(Q_e - Q_i)]^{1/2}} \right\}. \quad (46)$$

C. Viscous-Inertial Regime

In the viscous-inertial resonant response regime, Eqs. (19), (20), and (30) yield

$$\hat{\psi}_{0VI}(\hat{t}) = \frac{1}{2\pi i} \int_C \bar{\Psi}_{0VI}(\hat{g}) e^{\hat{g}\hat{t}} d\hat{g}, \quad (47)$$

where

$$\bar{\Psi}_{0VI}(\hat{g}) = \frac{\epsilon_{VI}}{\hat{g}} \frac{1}{[\hat{g} + i(Q_E + Q_e)]^{5/4}}, \quad (48)$$

and

$$\epsilon_{VI} = \frac{1}{\pi^{1/2} \Sigma P_\varphi^{1/4}}. \quad (49)$$

In this case,

$$\bar{H}(g) = \frac{1}{[\hat{g} + i(Q_E + Q_e)]^{5/4}}, \quad (50)$$

so Eq. (5.4.1) of Ref. 23 yields

$$H(\hat{t}) = \frac{1}{\Gamma(5/4)} \hat{t}^{1/4} e^{-i(Q_E + Q_e)\hat{t}}. \quad (51)$$

Hence,

$$\hat{\psi}_{0VI}(\hat{t}) = \frac{\epsilon_{VI}}{\Gamma(5/4)} \int_0^{\hat{t}} \hat{t}'^{1/4} e^{-i(Q_E + Q_e)\hat{t}'} d\hat{t}' \quad (52)$$

At small $\hat{t}$,

$$\hat{\psi}_{0VI}(\hat{t}) \simeq \frac{\epsilon_{VI} \hat{t}^{5/4}}{\Gamma(9/4)}. \quad (53)$$

At large $\hat{t}$,

$$\hat{\psi}_{0VI}(\hat{t}) \simeq \frac{\epsilon_{VI}}{\Gamma(5/4)} \int_0^\infty \hat{t}'^{1/4} e^{-i(Q_E + Q_e)\hat{t}'} d\hat{t}' \quad (54)$$

But Eq. (3.381.5) of Ref. 29 yields,

$$\int_0^\infty x^{\nu-1} e^{-iqx} dx = \frac{\Gamma(\nu)}{(iq)^\nu}, \quad (55)$$

so

$$\hat{\psi}_{0VI}(\hat{t}) \simeq \frac{\epsilon_{VI}}{[i(Q_E + Q_e)]^{5/4}}. \quad (56)$$

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Appendix A: Outer Region

1. Introduction

Consider the cylindrical plasma equilibrium introduced in Sect. II A. Suppose that the plasma is subject to an RMP with m periods in the poloidal direction, and n periods in the toroidal direction. The perturbation is resonant at the rational surface, radius r_s , at which $q(r_s) = m/n$. This appendix considers the response of the plasma to the RMP in the outer region: that is, everywhere apart from the immediate vicinity of the rational surface.

2. Cylindrical Tearing Mode Equation

In the outer region, the perturbed magnetic field can be written $\delta\mathbf{B} = r_s B_\varphi \nabla\delta\psi \times \mathbf{e}_z$, where

$$\delta\psi(r, \zeta, t) = \psi_1(r, t) e^{i\zeta}, \quad (\text{A1})$$

and $\zeta = m\theta - n\varphi$ is a helical angle.¹

The function $\psi_1(r, t)$ satisfies the *cylindrical tearing mode equation*,^{1,13,17,18}

$$\frac{\partial^2 \psi_1}{\partial \hat{r}^2} + \frac{1}{\hat{r}} \frac{\partial \psi_1}{\partial \hat{r}} - \frac{m^2}{\hat{r}^2} \psi_1 - \frac{J'_\varphi \psi_1}{\hat{r} (1/q - n/m)} = 0, \quad (\text{A2})$$

where $\hat{r} = r/a$, $J'_\varphi(\hat{r}) = dJ_\varphi/d\hat{r}$,

$$J_\varphi(\hat{r}) = \frac{R_0 \mu_0 j_\varphi}{B_\varphi}. \quad (\text{A3})$$

Let us assume that $J_\varphi(\hat{r})$ is a continuous function, and that $J_\varphi(\hat{r} > 1) = 0$. (In other words, the region $\hat{r} > 1$ is assumed to be a vacuum.) Note that Eq. (A2) is singular at the rational surface. The singularity is indicative of the breakdown of the equations of linearized, marginally-stable, ideal-MHD in the inner region.

3. Solution of Cylindrical Tearing Mode Equation

Suppose that the plasma is surrounded by a rigid conducting wall located at $r = b$, where $b > a$. Let $\psi_1(b, t) = \Psi_b(t)$, where $\Psi_b(t)$ is a complex dimensionless quantity that specifies the amplitude and phase of the RMP to which the plasma is subject. The most general solution to the cylindrical tearing mode equation in the outer region is written

$$\psi_1(\hat{r}, t) = \Psi_s(t) \hat{\psi}_s(\hat{r}) + \Psi_b(t) \hat{\psi}_b(\hat{r}), \quad (\text{A4})$$

Here, $\hat{\psi}_s(\hat{r})$ is a real continuous solution of Eq. (A2) that satisfies

$$\hat{\psi}_s(0) = 0, \quad (\text{A5})$$

$$\hat{\psi}_s(\hat{r}_s) = 1, \quad (\text{A6})$$

$$\hat{\psi}_s(\hat{r} \geq \hat{b}) = 0, \quad (\text{A7})$$

where $\hat{r}_s = r_s/a$, and $\hat{b} = b/a$. Moreover, $\hat{\psi}_b(\hat{r})$ is a real continuous solution of Eq. (A2) that satisfies

$$\hat{\psi}_b(\hat{r} \leq \hat{r}_s) = 0, \quad (\text{A8})$$

$$\hat{\psi}_b(\hat{b}) = 1. \quad (\text{A9})$$

Note that the dimensionless complex quantity $\Psi_s(t)$ parameterizes the amplitude and phase of the helical magnetic flux driven by the RMP at the interface between the inner and the outer regions.

4. Calculation of $\hat{\psi}_s(\hat{r})$

In the limit $0 < \hat{r} \ll 1$, the solution of Eq. (A2), subject to the boundary condition (A5), is

$$\hat{\psi}_s(\hat{r}) \propto \hat{r}^m + \kappa_0 \hat{r}^{m+2}, \quad (\text{A10})$$

where

$$\kappa_0 = -\frac{x_0}{4(m+1)(1/q_0 - n/m)}, \quad (\text{A11})$$

and $q_0 = q(0)$, with $x_0 = \lim_{\hat{r} \rightarrow 0} (-J'_\varphi/\hat{r})$.

Let $\hat{x} = (\hat{r} - \hat{r}_s)/\hat{r}_s$. In the immediate vicinity of the rational surface, the continuous solution to Eq. (A2), subject to the constraint (A6), is

$$\hat{\psi}_s(\hat{x}) = 1 + \Delta_{s+} \hat{x} + \alpha_s \hat{x} \ln |\hat{x}| + \mathcal{O}(\hat{x}^2) \quad (\text{A12})$$

for $\hat{x} > 0$, and

$$\hat{\psi}_s(\hat{x}) = 1 + \Delta_{s-} \hat{x} + \alpha_s \hat{x} \ln |\hat{x}| + \mathcal{O}(\hat{x}^2) \quad (\text{A13})$$

for $\hat{x} < 0$, where

$$\alpha_s = - \left(\frac{\hat{r} q J'_\varphi}{s} \right)_{\hat{r}=\hat{r}_s}, \quad (\text{A14})$$

and $s = d \ln q / d \ln r$ is the *magnetic shear*. In particular,

$$\Delta_{s\pm} = \frac{\hat{r}}{\hat{\psi}_s} \frac{d\hat{\psi}_s}{d\hat{r}} - \alpha_s (1 + \ln |\hat{x}|), \quad (\text{A15})$$

in the limit that $\hat{x} \rightarrow 0_{\pm}$.

Finally, solving Eq. (A2) outside the plasma (where $J'_\varphi = 0$), subject to the boundary condition (A7), yields

$$\left(\frac{\hat{r}}{\hat{\psi}_s} \frac{d\hat{\psi}_s}{d\hat{r}} \right)_{\hat{r}=1} = -m \left(\frac{\hat{b}^{2m} + 1}{\hat{b}^{2m} - 1} \right). \quad (\text{A16})$$

We can calculate $\hat{\psi}_s(\hat{r})$ by, first, launching a solution of Eq. (A2) from the magnetic axis, subject to the boundary condition (A10), and integrating to just inside the rational surface. Next, we launch a second solution of Eq. (A2) from the plasma boundary, subject to the boundary condition (A16), and integrate to just outside the rational surface. The two solutions are then rescaled subject to the constraint $\hat{\psi}_s(\hat{r}_s) = 1$. The real dimensionless quantity

$$E_{ss} \equiv \left[\hat{r} \frac{d\hat{\psi}_s}{d\hat{r}} \right]_{\hat{r}=\hat{r}_{s-}}^{\hat{r}=\hat{r}_{s+}} = \Delta_{s+} - \Delta_{s-} \quad (\text{A17})$$

is the conventional tearing stability index.¹⁴ It is assumed that $E_{ss} < 0$, which implies that the m/n tearing mode is intrinsically stable. Thus, any reconnected magnetic flux that develops at the rational surface is due solely to the action of the RMP to which the plasma is subject.

5. Calculation of $\hat{\psi}_b(\hat{r})$

In order to calculate $\hat{\psi}_b(\hat{r})$, we launch a solution of Eq. (A2) from just outside the rational surface, such that

$$\hat{\psi}_b(\hat{r}_s) = 0, \quad (\text{A18})$$

$$\frac{d\hat{\psi}_b(\hat{r}_s)}{d\hat{r}} = \frac{1}{\hat{r}_s}, \quad (\text{A19})$$

and then integrate it to the plasma boundary. Let

$$\alpha_b = \frac{1}{2m} \left[m \hat{b}^m \hat{\psi}_b(1) + \hat{b}^m \frac{d\hat{\psi}_b(1)}{d\hat{r}} \right], \quad (\text{A20})$$

$$\beta_b = \frac{1}{2m} \left[m \hat{b}^{-m} \hat{\psi}_b(1) - \hat{b}^{-m} \frac{d\hat{\psi}_b(1)}{d\hat{r}} \right], \quad (\text{A21})$$

$$\gamma_b = \alpha_b + \beta_b. \quad (\text{A22})$$

The boundary condition (A9) is satisfied by dividing our solution by γ_b . Let us define the real dimensionless quantity

$$E_{sb} \equiv \left[\hat{r} \frac{d\hat{\psi}_b}{d\hat{r}} \right]_{\hat{r}=\hat{r}_{s+}} = \frac{1}{\gamma_b}, \quad (\text{A23})$$

which parameterizes the coupling between the wall and the rational surface.

6. Asymptotic Matching

The complex dimensionless quantity

$$\Delta\Psi_s(t) = \left[\hat{r} \frac{\partial\psi_1}{\partial\hat{r}} \right]_{\hat{r}=\hat{r}_{s-}}^{\hat{r}=\hat{r}_{s+}} \quad (\text{A24})$$

parameterizes the amplitude and phase of the helical current sheet that is induced in the inner region. Asymptotic matching between the inner and the outer regions yields

$$\Delta\Psi_s(t) = E_{ss} \Psi_s(t) + E_{sb} \Psi_b(t), \quad (\text{A25})$$

where use has been made of Eqs. (A4), (A17), (A23), and (A24).

Appendix B: Inner Region

1. Introduction

The inner region, which lies in the immediate vicinity of the rational surface, is assumed to be governed by the four-field drift-MHD model described in Ref. 16. This model is an extension of a model introduced in Ref. 15, and is ultimately based on the four-field model of Ref. 19.

2. Useful Definitions

The plasma is assumed to consist of two species. First, electrons of mass m_e , electrical charge $-e$, number density n_e , and temperature T_e . Second, ions of mass m_i , electrical charge $+e$, number density n_e , and temperature T_i . Let $p = n_e (T_e + T_i)$ be the total plasma pressure.

It is helpful to define $n_0 = n_e(r_s)$, $p_0 = p(r_s)$,

$$\eta_e = \left. \frac{d \ln T_e}{d \ln n_e} \right|_{r=r_s}, \quad (\text{B1})$$

$$\eta_i = \left. \frac{d \ln T_i}{d \ln n_e} \right|_{r=r_s}, \quad (\text{B2})$$

$$\iota \equiv \left(\frac{dp_e}{dp_i} \right)_{r_s} = \left(\frac{T_e}{T_i} \right)_{r=r_s} \left(\frac{1 + \eta_e}{1 + \eta_i} \right), \quad (\text{B3})$$

where $n_e(r)$, $p(r)$, $p_e(r) = n_e(r) T_e(r)$, $p_i(r) = n_e(r) T_i(r)$, $T_e(r)$, and $T_i(r)$ refer to density, total pressure, electron pressure, ion pressure, electron temperature, and ion temperature profiles, respectively, in the absence of the RMP.

For the sake of simplicity, the perturbed electron and ion temperature profiles are assumed to be functions of the perturbed electron number density profile in the immediate vicinity of the rational surface. In other words, $T_e = T_e(n_e)$ and $T_i = T_i(n_e)$. This implies that $p = p(n_e)$. The MHD velocity, which is the velocity of a fictional MHD fluid,³⁰ is defined as $\mathbf{V} = \mathbf{V}_E + V_{\parallel i} \mathbf{b}$, where $\mathbf{V}_E$ is the $\mathbf{E} \times \mathbf{B}$ drift velocity, $V_{\parallel i}$ the parallel component of the ion fluid velocity, $\mathbf{b} = \mathbf{B}/|\mathbf{B}|$, and $\mathbf{B}$ the magnetic field-strength.

3. Dimensionless Fields

The four dimensionless fields in our four-field model—namely, ψ , ϕ , N , and V —have the following definitions:

$$\nabla\psi = \frac{\mathbf{e}_{\parallel} \times \mathbf{B}}{r_s B_{\varphi}}, \quad (\text{B4})$$

$$\nabla\phi = \frac{\mathbf{e}_{\parallel} \times \mathbf{V}}{r_s V_A}, \quad (\text{B5})$$

$$N = -\hat{d}_i \left(\frac{p - p_0}{B_{\varphi}^2 / \mu_0} \right), \quad (\text{B6})$$

$$V = \hat{d}_i \left(\frac{\mathbf{e}_{\parallel} \cdot \mathbf{V}}{V_A} \right). \quad (\text{B7})$$

Here, $\mathbf{e}_{\parallel} = (0, \epsilon/q_s, 1)$, $\epsilon = r/R_0$, $q_s = m/n$, $V_A = (B_{\varphi}/\sqrt{\mu_0 n_0 m_i})_{r_s}$ is the Alfvén speed, $d_i = \sqrt{m_i/(n_0 e^2 \mu_0)_{r_s}}$ the collisionless ion skin-depth, and $\hat{d}_i = d_i/r_s$. Our model also employs the auxiliary dimensionless field

$$J = -\frac{2\epsilon_s}{q_s} + \hat{\nabla}^2 \psi, \quad (\text{B8})$$

where $\epsilon_s = r_s/R_0$, and $\hat{\nabla} = r_s \nabla$. Note that

$$\psi(r, \zeta, t) = \psi_0(r) + \psi_1(r, t) e^{i\zeta}, \quad (\text{B9})$$

where

$$\psi_0(r) = \frac{1}{R_0 r_s} \int_{r_s}^r r' \left[\frac{n}{m} - \frac{1}{q(r')} \right] dr'. \quad (\text{B10})$$

4. Four-Field Model

Our four-field model takes the form:^{15,16,20}

$$\frac{\partial \psi}{\partial \hat{t}} = [\phi, \psi] - \iota_e [N, \psi] + \hat{\eta}_{\parallel} J + \hat{E}_{\parallel}, \quad (\text{B11})$$

$$\begin{aligned} \frac{\partial \hat{\nabla}^2 \phi}{\partial \hat{t}} &= [\phi, \hat{\nabla}^2 \phi] - \frac{\iota_i}{2} \left(\hat{\nabla}^2 [\phi, N] + [\hat{\nabla}^2 \phi, N] + [\hat{\nabla}^2 N, \phi] \right) + [J, \psi] \\ &\quad + \hat{\chi}_{\varphi} \hat{\nabla}^4 (\phi + \iota_i N), \end{aligned} \quad (\text{B12})$$

$$\frac{\partial N}{\partial \hat{t}} = [\phi, N] + c_{\beta}^2 [V, \psi] + \hat{d}_{\beta}^2 [J, \psi] + \hat{\chi}_{\perp} \hat{\nabla}^2 N, \quad (\text{B13})$$

$$\frac{\partial V}{\partial \hat{t}} = [\phi, V] + [N, \psi] + \hat{\chi}_{\varphi} \hat{\nabla}^2 V. \quad (\text{B14})$$

Here, $[A, B] \equiv \hat{\nabla} A \times \hat{\nabla} B \cdot \mathbf{e}_{\parallel}$, $\iota_e = \iota/(1 + \iota)$, $\iota_i = 1/(1 + \iota)$, $\hat{t} = t/(r_s/V_A)$, $\hat{\eta}_{\parallel} = \eta_{\parallel}/(\mu_0 r_s V_A)$, $\hat{E}_{\parallel} = E_{\parallel}/(B_{\varphi} V_A)$, $\hat{\chi}_{\varphi} = \chi_{\varphi}/(r_s V_A)$, $\hat{\chi}_{\perp} = \chi_{\perp}/(r_s V_A)$, where $\eta_{\parallel}$ is the parallel plasma electrical resistivity at the rational surface, $E_{\parallel}$ the parallel inductive electric field that maintains the equilibrium toroidal plasma current in the vicinity of the rational surface, χ_{φ} the anomalous perpendicular ion momentum diffusivity at the rational surface, and $\chi_{\perp}$ the anomalous perpendicular energy diffusivity at the rational surface. Moreover, $d_{\beta} = c_{\beta} d_i$, and $\hat{d}_{\beta} = d_{\beta}/r_s$, where $c_{\beta} = \sqrt{\beta/(1 + \beta)}$, and $\beta = (5/3) \mu_0 p_0/B_{\varphi}^2$. Here, d_{β} is usually referred to as the *ion sound-radius*. Note that we have neglected parallel energy transport in Eq. (B13) because we expect the radial width of the inner region to be less than the critical width above which parallel transport forces the temperature and density profiles to be magnetic flux-surface functions.²¹

5. Matching to Plasma Equilibrium

The unperturbed plasma equilibrium is such that $\mathbf{B} = (0, B_{\theta}(r), B_{\varphi})$, $p = p(r)$, $\mathbf{V} = (0, V_E(r), V_{\varphi}(r))$, where $V_E(r) \simeq -E_r/B_{\varphi}$ is the (dominant θ -component of the) $\mathbf{E} \times \mathbf{B}$ drift velocity. Now, the inner region is assumed to have a radial thickness that is much smaller than r_s . Hence, we only need to evaluate plasma equilibrium quantities in the immediate vicinity of the rational surface. Equations (B4)–(B7) and (B9)–(B10) suggest that

$$\psi(\hat{x}) = \frac{\hat{x}^2}{2 \hat{L}_s}, \quad (\text{B15})$$

$$\phi(\hat{x}) = -\hat{V}_E \hat{x}, \quad (\text{B16})$$

$$N(\hat{x}) = -\hat{V}_* \hat{x}, \quad (\text{B17})$$

$$V = \hat{V}_{\parallel}, \quad (\text{B18})$$

where $\hat{x} = (r - r_s)/r_s$, $\hat{L}_s = L_s/r_s$, $L_s = R_0 q_s/s_s$, $\hat{V}_E = V_E(r_s)/V_A$, $\hat{V}_* = V_*(r_s)/V_A$, $V_*(r) = (dp/dr)/(e n_0 B_{\varphi})$ is the (dominant θ -component of the) diamagnetic velocity, and $\hat{V}_{\parallel} = \hat{d}_i V_{\varphi}(r_s)/V_A$. Here, $s_s = s(r_s)$. We also deduce that

$$J(\hat{x}) = - \left(\frac{2}{s_s} - 1 \right) \frac{1}{\hat{L}_s}, \quad (\text{B19})$$

and $\hat{E}_{\parallel}(\hat{x}) = (2/s_s - 1)(\hat{\eta}_{\parallel}/\hat{L}_s)$.

6. Linearized Layer Equations

In accordance with Eqs. (B8), (B9), and (B15)–(B19), let us write

$$\psi(\hat{x}, \zeta, \hat{t}) = \frac{\hat{x}^2}{2 \hat{L}_s} + \tilde{\psi}(\hat{x}, \hat{t}) e^{i\zeta}, \quad (\text{B20})$$

$$\phi(\hat{x}, \zeta, \hat{t}) = -\hat{V}_E \hat{x} + \tilde{\phi}(\hat{x}, \hat{t}) e^{i\zeta}, \quad (\text{B21})$$

$$N(\hat{x}, \zeta, \hat{t}) = -\hat{V}_* \hat{x} + \iota_e \tilde{N}(\hat{x}, \hat{t}) e^{i\zeta}, \quad (\text{B22})$$

$$V(\hat{x}, \zeta, \hat{t}) = \hat{V}_\parallel + \iota_e \tilde{V}(\hat{x}, \hat{t}) e^{i\zeta}, \quad (\text{B23})$$

$$J(\hat{x}, \zeta, \hat{t}) = -\left(\frac{2}{s_s} - 1\right) \frac{1}{\hat{L}_s} + \hat{\nabla}^2 \tilde{\psi}(\hat{x}, \hat{t}) e^{i\zeta}. \quad (\text{B24})$$

Substituting Eqs. (B20)–(B24) into Eqs. (B11)–(B14), and only retaining terms that are first order in perturbed quantities, we obtain the following set of linearized layer equations:

$$\tau_H \frac{\partial \tilde{\psi}}{\partial t} - i n (\omega_E + \omega_{*e}) \tau_H \tilde{\psi} = -i \hat{x} (\tilde{\phi} - \tilde{N}) + S^{-1} \hat{\nabla}^2 \tilde{\psi}, \quad (\text{B25})$$

$$\tau_H \frac{\partial \hat{\nabla}^2 \tilde{\phi}}{\partial t} - i n (\omega_E + \omega_{*i}) \tau_H \hat{\nabla}^2 \tilde{\phi} = -i \hat{x} \hat{\nabla}^2 \tilde{\psi} + S^{-1} P_\varphi \hat{\nabla}^4 \left(\tilde{\phi} + \frac{\tilde{N}}{\iota} \right), \quad (\text{B26})$$

$$\begin{aligned} \tau_H \frac{\partial \tilde{N}}{\partial t} - i n \omega_E \tau_H \tilde{N} &= i n \omega_{*e} \tau_H \tilde{\phi} - i c_\beta^2 \hat{x} \tilde{V} - i \iota_e \hat{d}_\beta^2 \hat{x} \hat{\nabla}^2 \tilde{\psi} \\ &+ S^{-1} P_\perp \hat{\nabla}^2 \tilde{N}, \end{aligned} \quad (\text{B27})$$

$$\tau_H \frac{\partial \tilde{V}}{\partial t} - i n \omega_E \tau_H \tilde{V} = -i n \omega_{*e} \tau_H \tilde{\psi} - i \hat{x} \tilde{N} + S^{-1} P_\varphi \hat{\nabla}^2 \tilde{V}. \quad (\text{B28})$$

Here, $\tau_H = L_s / (m V_A)$ is the *hydromagnetic* time, $\omega_E = -(q_s / r_s) V_E(r_s) = -(d\Phi / d\psi_p)_{r_s}$ the $\mathbf{E} \times \mathbf{B}$ frequency, $\omega_{*e} = \iota_e (q_s / r_s) V_*(r_s) = [(dp_e / d\psi_p) / (e n_e)]_{r_s}$ the electron diamagnetic frequency, $\omega_{*i} = -\iota_i (q_s / r_s) V_*(r_s) = -[(dp_i / d\psi_p) / (e n_e)]_{r_s}$ the ion diamagnetic frequency, $\Phi(r)$ the equilibrium electrostatic potential, $\psi_p(r) = B_\varphi \int_0^r dr' r' / q(r')$ the equilibrium poloidal magnetic flux, $S = \tau_R / \tau_H$ the Lundquist number, $\tau_R = (\mu_0 r_s^2 / \eta_\parallel)_{r_s}$ the resistive diffusion time, $\tau_\varphi = (r^2 / \chi_\varphi)_{r_s}$ the toroidal momentum confinement time, and $\tau_\perp = (r^2 / \chi_\perp)_{r_s}$ the energy confinement time. Furthermore, $P_\varphi = \tau_R / \tau_\varphi$ and $P_\perp = \tau_R / \tau_\perp$ are magnetic Prandtl numbers. Note that $\iota = -\omega_{*e} / \omega_{*i}$.

In the following, in accordance with the analysis of Refs. 15 and 16, we shall neglect the term $-i c_\beta^2 \hat{x} \tilde{V}$ in Eq. (B27). This approximation is only valid in low- β plasmas, and is equivalent to the neglect of ion parallel dynamics in the inner region. The low- β approximation

effectively converts our four-field model into the following three-field model:

$$\tau_H \frac{\partial \tilde{\psi}}{\partial t} - i n (\omega_E + \omega_{*e}) \tau_H \tilde{\psi} = -i \hat{x} (\tilde{\phi} - \tilde{N}) + S^{-1} \hat{\nabla}^2 \tilde{\psi}, \quad (\text{B29})$$

$$\tau_H \frac{\partial \hat{\nabla}^2 \tilde{\phi}}{\partial t} - i n (\omega_E + \omega_{*i}) \tau_H \hat{\nabla}^2 \tilde{\phi} = -i \hat{x} \hat{\nabla}^2 \tilde{\psi} + S^{-1} P_\varphi \hat{\nabla}^4 \left(\tilde{\phi} + \frac{\tilde{N}}{\iota} \right), \quad (\text{B30})$$

$$\tau_H \frac{\partial \tilde{N}}{\partial t} - i n \omega_E \tau_H \tilde{N} = i n \omega_{*e} \tau_H \tilde{\phi} - i \iota_e \hat{d}_\beta^2 \hat{x} \hat{\nabla}^2 \tilde{\psi} + S^{-1} P_\perp \hat{\nabla}^2 \tilde{N}. \quad (\text{B31})$$

7. Laplace Transformed Layer Equations

Let us Laplace transform Eqs. (B29)–(B31) by operating on them with $\int_0^\infty (\dots) e^{-gt} dt$. Assuming that all fields are zero at $t = 0$, we obtain

$$g \tau_H \bar{\psi} - i n (\omega_E + \omega_{*e}) \tau_H \bar{\psi} = -i \hat{x} (\bar{\phi} - \bar{N}) + S^{-1} \hat{\nabla}^2 \bar{\psi}, \quad (\text{B32})$$

$$g \tau_H \hat{\nabla}^2 \bar{\phi} - i n (\omega_E + \omega_{*i}) \tau_H \hat{\nabla}^2 \bar{\phi} = -i \hat{x} \hat{\nabla}^2 \bar{\psi} + S^{-1} P_\varphi \hat{\nabla}^4 \left(\bar{\phi} + \frac{\bar{N}}{\iota} \right), \quad (\text{B33})$$

$$g \tau_H \bar{N} - i n \omega_E \tau_H \bar{N} = i n \omega_{*e} \tau_H \bar{\phi} - i \iota_e \hat{d}_\beta^2 \hat{x} \hat{\nabla}^2 \bar{\psi} + S^{-1} P_\perp \hat{\nabla}^2 \bar{N}, \quad (\text{B34})$$

where

$$\bar{\psi}(\hat{x}, g) = \int_0^\infty \tilde{\psi}(\hat{x}, t) e^{-gt} dt, \quad (\text{B35})$$

et cetera.

8. Asymptotic Matching

Let

$$\bar{\Psi}_b(g) = \int_0^\infty \Psi_b(t) e^{-gt} dt, \quad (\text{B36})$$

$$\bar{\Psi}_s(g) = \int_0^\infty \Psi_s(t) e^{-gt} dt, \quad (\text{B37})$$

$$\overline{\Delta \Psi}_s(g) = \int_0^\infty \Delta \Psi_s(t) e^{-gt} dt \equiv \bar{\Delta}_s(g) \bar{\Psi}_s(g). \quad (\text{B38})$$

It follows from Eq. (A25) that

$$\bar{\Psi}_s(g) = \frac{E_{sb} \bar{\Psi}_b(g)}{\bar{\Delta}_s(g) - E_{ss}}. \quad (\text{B39})$$

The Laplace transformed layer equations, (B32)–(B34), possess a twisting parity solution,²² such that $\bar{\psi}(-\hat{x}, g) = -\bar{\psi}(\hat{x}, g)$, $\bar{\phi}(-\hat{x}, g) = \bar{\phi}(\hat{x}, g)$, and $\bar{N}(-\hat{x}, g) = \bar{N}(\hat{x}, g)$. In fact, an inspection of Eqs. (B32)–(B34) reveals that the twisting parity solution is trivial. Furthermore, when this solution is asymptotically matched to the outer region, making use of Eqs. (A4), (A12), (A13), and (A23), we obtain

$$\bar{\psi}(\hat{x}, g) = \frac{\bar{\xi}_r(g)}{L_s} \hat{x}, \quad (\text{B40})$$

$$\bar{\phi}(\hat{x}, g) = \text{i} (g - \text{i} n \omega_E) \tau_H \frac{\bar{\xi}_r(g)}{L_s}, \quad (\text{B41})$$

$$\bar{N}(\hat{x}, g) = -n \omega_{*e} \tau_H \frac{\bar{\xi}_r(g)}{L_s}. \quad (\text{B42})$$

Here,

$$\bar{\xi}_r(g) = \int_0^\infty \xi_r(t) e^{-gt} dt, \quad (\text{B43})$$

$$\xi_r(t) = \frac{(\Delta_{s+} + \Delta_{s-}) \Psi_s(t) + E_{sb} \Psi_b(t)}{2/L_s}, \quad (\text{B44})$$

where $\xi_r(t) e^{\text{i}\zeta}$ is an even (in $\hat{x}$) radial displacement of the plasma in the vicinity of the rational surface that turns out to be independent of $\hat{x}$, and is driven by the magnetic perturbation. It follows that²³

$$\tilde{\psi}(\hat{x}, t) = \frac{\xi_r(t)}{L_s} \hat{x}, \quad (\text{B45})$$

$$\tilde{\phi}(t) = \text{i} \frac{\tau_H}{L_s} \left(\frac{d}{dt} - \text{i} n \omega_E \right) \xi_r(t), \quad (\text{B46})$$

$$\tilde{N}(t) = -n \omega_{*e} \tau_H \frac{\xi_r(t)}{L_s}. \quad (\text{B47})$$

Note that the twisting parity solution gives rise to no magnetic reconnection at the rational surface [i.e., $\tilde{\psi}(\hat{x} = 0, \zeta, \hat{t}) = 0$].

The Laplace transformed layer equations, (B32)–(B34), also possess a non-trivial tearing parity solution,²² such that $\bar{\psi}(-\hat{x}, g) = \bar{\psi}(\hat{x}, g)$, $\bar{\phi}(-\hat{x}, g) = -\bar{\phi}(\hat{x}, g)$, and $\bar{N}(-\hat{x}, g) = -\bar{N}(\hat{x}, g)$. When this solution is asymptotically matched to the outer solution, making use of Eqs. (A4), (A12), (A13), and (B4), we find that

$$\bar{\psi}(\hat{x}, g) \rightarrow \bar{\Psi}_s(g) \left[1 + \bar{\Delta}_s(g) \frac{|\hat{x}|}{2} \right] \quad (\text{B48})$$

at the interface between the inner and the outer regions.

9. Stretched Radial Coordinate

Let us define the stretched radial coordinate $X = S^{1/3} \hat{x}$. Assuming that $X \sim \mathcal{O}(1)$ in the resonant layer that constitutes the inner region (i.e., assuming that the thickness of the layer is roughly of order $S^{-1/3} r_s$), and making use of the fact that $S \gg 1$ in a conventional tokamak plasma, Eqs. (B32)–(B34) reduce to the following set of linear layer equations:

$$[\hat{g} + \mathrm{i}(Q_E + Q_e)] \bar{\psi} = -\mathrm{i} X (\bar{\phi} - \bar{N}) + \frac{\partial^2 \bar{\psi}}{\partial X^2}, \quad (\text{B49})$$

$$[\hat{g} + \mathrm{i}(Q_E + Q_i)] \frac{\partial^2 \bar{\phi}}{\partial X^2} = -\mathrm{i} X \frac{\partial^2 \bar{\psi}}{\partial X^2} + P_\varphi \frac{\partial^4}{\partial X^4} \left(\bar{\phi} + \frac{\bar{N}}{\iota} \right), \quad (\text{B50})$$

$$(\hat{g} + \mathrm{i} Q_E) \bar{N} = -\mathrm{i} Q_e \bar{\phi} - \mathrm{i} D^2 X \frac{\partial^2 \bar{\psi}}{\partial X^2} + P_\perp \frac{\partial^2 \bar{N}}{\partial X^2}, \quad (\text{B51})$$

where

$$\hat{g} = S^{1/3} g \tau_H, \quad (\text{B52})$$

$$Q_E = -S^{1/3} n \omega_E \tau_H, \quad (\text{B53})$$

$$Q_{e,i} = -S^{1/3} n \omega_{*e,i} \tau_H, \quad (\text{B54})$$

$$D = S^{1/3} \iota_e^{1/2} \hat{d}_\beta. \quad (\text{B55})$$

According to Eq. (B48), the asymptotic behavior of the tearing-parity solutions of Eqs. (B49)–(B51) is such that

$$\bar{\psi}(X, \hat{g}) \rightarrow \bar{\Psi}_s(g) \left[\frac{\hat{\Delta}_s(\hat{g})}{2} |X| + 1 + \mathcal{O}\left(\frac{1}{X^2}\right) \right] \quad (\text{B56})$$

as $|X| \rightarrow \infty$, where

$$\bar{\Delta}_s(g) = S^{1/3} \hat{\Delta}_s(\hat{g}). \quad (\text{B57})$$

It follows from Eqs. (B49)–(B51) that

$$\bar{\phi}(X, \hat{g}) \rightarrow \mathrm{i} (\hat{g} + \mathrm{i} Q_E) \bar{\Psi}_s(g) \left[\frac{\hat{\Delta}_s(\hat{g})}{2} \operatorname{sgn}(X) + \frac{1}{X} + \mathcal{O}\left(\frac{1}{X^2}\right) \right] \quad (\text{B58})$$

as $|X| \rightarrow \infty$.

10. Fourier Transformation

Equations (B49)–(B51) are most conveniently solved in Fourier transform space.¹⁵ Let

$$\hat{\phi}(p, \hat{g}) = \int_{-\infty}^{\infty} \bar{\phi}(X, \hat{g}) e^{-\mathrm{i} p X} dX, \quad (\text{B59})$$

et cetera. The Fourier transformed linear layer equations become

$$[\hat{g} + \mathbf{i}(Q_E + Q_e)] \hat{\psi} = \frac{\partial}{\partial p} \left(\hat{\phi} - \hat{N} \right) - p^2 \hat{\psi}, \quad (\text{B60})$$

$$[\hat{g} + \mathbf{i}(Q_E + Q_i)] p^2 \hat{\phi} = \frac{\partial(p^2 \hat{\psi})}{\partial p} - P_\varphi p^4 \left(\hat{\phi} + \frac{\hat{N}}{\iota} \right), \quad (\text{B61})$$

$$(\hat{g} + \mathbf{i} Q_E) \hat{N} = -\mathbf{i} Q_e \hat{\phi} - D^2 \frac{\partial(p^2 \hat{\psi})}{\partial p} - P_\perp p^2 \hat{N}, \quad (\text{B62})$$

where, for a tearing parity solution, Eq. (B58) yields²³

$$\hat{\phi}(p, \hat{g}) \rightarrow \pi (\hat{g} + \mathbf{i} Q_E) \bar{\Psi}_s(g) \left[\frac{\hat{\Delta}_s(\hat{g})}{\pi p} + 1 + \mathcal{O}(p) \right]. \quad (\text{B63})$$

as $p \rightarrow 0$.

Let

$$Y_e(p, g) \equiv \hat{\phi}(p, \hat{g}) - \hat{N}(p, \hat{g}) = \pi [\hat{g} + \mathbf{i}(Q_E + Q_e)] \bar{\Psi}_s(g) \hat{Y}_e(p, \hat{g}). \quad (\text{B64})$$

Equations (B60)–(B62) can be combined to give

$$\frac{\partial}{\partial p} \left[A(p, \hat{g}) \frac{\partial \hat{Y}_e}{\partial p} \right] - \frac{B(p, \hat{g})}{C(p, \hat{g})} p^2 \hat{Y}_e = 0, \quad (\text{B65})$$

where

$$A(p, \hat{g}) = \frac{p^2}{\hat{g} + \mathbf{i}(Q_E + Q_e) + p^2}, \quad (\text{B66})$$

$$B(p, \hat{g}) = (\hat{g} + \mathbf{i} Q_E) [\hat{g} + \mathbf{i}(Q_E + Q_i)] + [\hat{g} + \mathbf{i}(Q_E + Q_i)] (P_\varphi + P_\perp) p^2 + P_\varphi P_\perp p^4, \quad (\text{B67})$$

$$C(p, \hat{g}) = \hat{g} + \mathbf{i}(Q_E + Q_e) + \{P_\perp + [\hat{g} + \mathbf{i}(Q_E + Q_i)] D^2\} p^2 + \iota_e^{-1} P_\varphi D^2 p^4. \quad (\text{B68})$$

Furthermore, because

$$\hat{Y}_e(p, \hat{g}) = \left[\frac{\hat{g} + \mathbf{i}(Q_E + Q_e)}{\hat{g} + \mathbf{i} Q_E} \right] \hat{\phi}(p, \hat{g}) \quad (\text{B69})$$

as $p \rightarrow 0$, Eqs. (B63) and (B64) yield the following small- p boundary condition that Eq. (B65) must satisfy:²³

$$\hat{Y}_e(p, \hat{g}) \rightarrow \frac{\hat{\Delta}_s(\hat{g})}{\pi p} + 1 + \mathcal{O}(p) \quad (\text{B70})$$

as $p \rightarrow 0$. Equation (B65) must also satisfy the physical boundary condition

$$\hat{Y}_e(p, \hat{g}) \rightarrow 0 \quad (\text{B71})$$

as $p \rightarrow \infty$.

11. Reconnected Helical Magnetic Flux

The true reconnected helical magnetic flux at the rational surface (divided by $r_s B_\varphi$) is

$$\Psi_0(t) \equiv \psi_1(0, t) = \tilde{\psi}(0, t). \quad (\text{B72})$$

Let

$$\bar{\Psi}_0(g) = \int_0^\infty \Psi_0(t) e^{-gt} dt. \quad (\text{B73})$$

It is easily demonstrated from Eqs. (B49) and (B60) that ⁶

$$\frac{\bar{\Psi}_0(g)}{\bar{\Psi}_s(g)} \equiv F_s(\hat{g}) = - \int_0^\infty A(p, \hat{g}) \frac{\partial \hat{Y}_e}{\partial p} dp. \quad (\text{B74})$$

Thus, it follows from Eqs. (B39) and (B57) that

$$\bar{\Psi}_0(g) = \frac{E_{sb} F_s(\hat{g}) \bar{\Psi}_b(g)}{S^{1/3} \hat{\Delta}_s(\hat{g}) - E_{ss}}. \quad (\text{B75})$$

Appendix C: Linear Resonant Response Regimes

1. Introduction

The aim of this section is to classify all of the different linear resonant response regimes predicted by Eqs. (B65), (B70), (B71), and (B74). The approach taken is similar to that adopted in Refs. 15 and 16, except that we need to incorporate the Laplace transform parameter, g , into the analysis. In the following, we shall assume that $|Q_E| \sim |Q_e| \sim |Q_i| \sim Q$ and $P_\varphi \sim P_\perp \sim P$.

2. Constant- ψ Limit

Suppose that there are two layers in p -space. In the small- p layer, suppose that Eq. (B65) reduces to

$$\frac{\partial}{\partial p} \left[\frac{p^2}{\hat{g} + i(Q_E + Q_e) + p^2} \frac{\partial \hat{Y}_e}{\partial p} \right] \simeq 0 \quad (\text{C1})$$

when $p \sim R^{1/2}$, where $R = \sqrt{\hat{g}^2 + Q^2}$. Integrating directly, making use of Eq. (B70), we find that

$$\hat{Y}_e(p, g) \simeq \frac{\hat{\Delta}_s}{\pi} \left[\frac{1}{p} - \frac{p}{\hat{g} + i(Q_E + Q_e)} \right] + 1 + \mathcal{O}(p^2) \quad (\text{C2})$$

for $p \lesssim R^{1/2}$.

In the large- p limit, for $|p| \gg \mathcal{O}(R^{1/2})$, Eq. (B65) reduces to

$$\frac{\partial^2 \hat{Y}_e}{\partial p^2} - \frac{B(p, \hat{g})}{C(p, \hat{g})} p^2 \hat{Y}_e \simeq 0, \quad (\text{C3})$$

with $\hat{Y}_e(p, \hat{g})$ bounded as $p \rightarrow \infty$, in accordance with Eq. (B71). Asymptotic matching to the small- p layer solution, (C2), yields the following small- p boundary condition that the previous equation must satisfy:

$$\hat{Y}_e(p, \hat{g}) \simeq 1 - \frac{\hat{\Delta}_s}{\pi} \frac{p}{\hat{g} + i(Q_E + Q_e)} + \mathcal{O}(p^2) \quad (\text{C4})$$

as $p \rightarrow 0$.

In the various constant- ψ linear response regimes considered in Sect. C3, Eq. (C3) reduces to an equation of the form

$$\frac{\partial^2 \hat{Y}_e}{\partial p^2} - G p^k \hat{Y}_e \simeq 0, \quad (\text{C5})$$

where k is real and non-negative, and $G(\hat{g})$ is a complex constant. Let $\hat{Y}_e(p, \hat{g}) = \sqrt{p} Z(q)$ and $q = \sqrt{G} p^j / j$, where $j = (k + 2)/2$. The previous equation transforms into a modified Bessel equation of general order,

$$q^2 \frac{d^2 Z}{dq^2} + q \frac{dZ}{dq} - (q^2 + \nu^2) Z = 0, \quad (\text{C6})$$

where $\nu = 1/(k + 2)$. The solution that is bounded as $q \rightarrow \infty$ has the small- q expansion²⁴

$$K_\nu(q) = \frac{1}{\Gamma(1 - \nu)} \left(\frac{2}{q} \right)^\nu - \frac{1}{\Gamma(1 + \nu)} \left(\frac{q}{2} \right)^\nu + \mathcal{O}(q^{2-\nu}), \quad (\text{C7})$$

where $\Gamma(z)$ is a gamma function. A comparison of the previous expression with Eq. (C4) reveals that

$$\hat{\Delta}_s(\hat{g}) = \frac{\nu^{2\nu-1} \pi \Gamma(1 - \nu)}{\Gamma(\nu)} [\hat{g} + i(Q_E + Q_e)] G^\nu. \quad (\text{C8})$$

Note, finally, that $p_* \sim |G|^{-\nu}$, where p_* denotes the width of the large- p layer in p -space. This width must be larger than $R^{1/2}$ (i.e., the width of the small- p layer) in order for the constant- ψ approximation to hold. Finally, it is easily demonstrated from Eqs. (B71), (B74), (C4), and the fact that $p_* \gg R^{1/2}$, that

$$F_s(\hat{g}) \simeq 1 \quad (\text{C9})$$

in a constant- ψ layer. Thus, as the name suggests, a constant- ψ layer is characterized by a helical magnetic flux at the center of the layer, $\Psi_0(t)$, that is equal to that at the edge of the layer, $\Psi_s(t)$.

3. Constant- ψ Response Regimes

Suppose that $R \gg P p^2$ and $D^2 p^2 \ll 1$. It follows that $k = 2$, $j = 2$, $\nu = 1/4$, and

$$G(\hat{g}) = \frac{(\hat{g} + i Q_E) [\hat{g} + i (Q_E + Q_i)]}{\hat{g} + i (Q_E + Q_e)}. \quad (\text{C10})$$

Hence, we deduce that

$$\hat{\Delta}_s(\hat{g}) = \frac{2\pi \Gamma(3/4)}{\Gamma(1/4)} [\hat{g} + i (Q_E + Q_e)]^{3/4} (\hat{g} + i Q_E)^{1/4} [\hat{g} + i (Q_E + Q_i)]^{1/4}. \quad (\text{C11})$$

This response regime is known as the *resistive-inertial* because the layer response is dominated by plasma resistivity and ion inertia.^{2,14,25} The characteristic layer width is $p_* \sim R^{-1/4}$, which implies that the regime is valid when $P \ll R^{3/2}$, $R \gg D^4$, and $R \ll 1$.

Suppose that $R \ll P p^2$ and $D^2 p^2 \ll 1$. It follows that $k = 4$, $j = 3$, $\nu = 1/6$, and $G(\hat{g}) = P_\varphi$. Hence, we deduce that

$$\hat{\Delta}_s(\hat{g}) = \frac{6^{2/3} \pi \Gamma(5/6)}{\Gamma(1/6)} [\hat{g} + i (Q_E + Q_e)] P_\varphi^{1/6}. \quad (\text{C12})$$

This response regime is known as the *viscous-resistive* regime because the layer response is dominated by ion perpendicular viscosity and plasma resistivity.^{1,26} The characteristic layer width is $p_* \sim P^{-1/6}$, which implies that the regime is valid when $P \gg R^{3/2}$, $P \gg D^6$, and $P \ll R^{-3}$.

Suppose that $R \gg P p^2$ and $D^2 p^2 \gg 1$. It follows that $k = 0$, $j = 1$, $\nu = 1/2$, and

$$G(\hat{g}) = \frac{\hat{g} + i Q_E}{D^2}. \quad (\text{C13})$$

Hence, we deduce that

$$\hat{\Delta}_s(\hat{g}) = \frac{\pi [\hat{g} + i (Q_E + Q_e)] (\hat{g} + i Q_E)^{1/2}}{D}. \quad (\text{C14})$$

This response regime is known as the *semi-collisional* regime.^{27,28} The characteristic layer width is $p_* \sim D R^{-1/2}$, which implies that the regime is valid when $R \ll D^4$, $P \ll R^2/D^2$, and $R \ll D$.

Suppose, finally, that $R \ll P p^2$ and $D^2 p^2 \gg 1$. It follows that $k = 2$, $j = 2$, $\nu = 1/4$, and

$$G(\hat{g}) = \frac{\iota_e P_\perp}{D^2}. \quad (\text{C15})$$

Hence, we deduce that

$$\hat{\Delta}_s(\hat{g}) = \frac{2\pi \Gamma(3/4)}{\Gamma(1/4)} \frac{[\hat{g} + i(Q_E + Q_e)] \iota_e^{1/4} P_\perp^{1/4}}{D^{1/2}}. \quad (\text{C16})$$

This response regime is known as the *diffusive-resistive* regime because the layer response is dominated by perpendicular energy diffusivity and plasma resistivity.¹⁶ The characteristic layer width is $p_* \sim P^{-1/4} D^{1/2}$, which implies that the regime is valid when $P \gg R^2/D^2$, $P \ll D^6$, and $P \ll D^2/R^2$.

4. Non-constant- ψ Limit

Suppose that $p \ll R^{1/2}$. In this limit, Eq. (B65) reduces to

$$\frac{\partial}{\partial p} \left(p^2 \frac{\partial \hat{Y}_e}{\partial p} \right) - [\hat{g} + i(Q_E + Q_e)] \frac{B(p, \hat{g})}{C(p, \hat{g})} p^2 \hat{Y}_e = 0. \quad (\text{C17})$$

In the various non-constant- ψ regimes considered in Section C 5, the previous equation takes the form

$$\frac{\partial}{\partial p} \left(p^2 \frac{\partial \hat{Y}_e}{\partial p} \right) - G p^{k+2} \hat{Y}_e = 0, \quad (\text{C18})$$

where k is real and non-negative, and $G(\hat{g})$ is a complex constant. Let $V(p, \hat{g}) = p \hat{Y}_e(p, \hat{g})$.

The previous equation yields

$$\frac{\partial^2 V}{\partial p^2} - G p^k V = 0. \quad (\text{C19})$$

This equation is identical in form to Equation (C5), which we have already solved. Indeed, the solution that is bounded as $p \rightarrow \infty$ has the small- p expansion (C7), where $q = \sqrt{G} p^j / j$, $j = (k + 2)/2$, and $\nu = 1/(k + 2)$. Matching to Equation (B70) yields

$$\hat{\Delta}_s(\hat{g}) = -\frac{\pi \nu^{1-2\nu} \Gamma(\nu)}{\Gamma(1-\nu)} G^{-\nu}. \quad (\text{C20})$$

The layer width in p -space again scales as $p_* \sim |G|^{-\nu}$. This width must be less than $R^{1/2}$. Finally, it follows from Eqs. (B66) and (B74) that

$$F_s(\hat{g}) = \frac{2}{\hat{g} + i(Q_E + Q_e)} \int_0^\infty V(p, \hat{g}) dp \quad (\text{C21})$$

in a non-constant- ψ response regime.

5. Non-constant- ψ Linear Resonant Response Regimes

Suppose that $R \gg P p^2$ and $D^2 p^2 \ll 1$. It follows that $k = 0$, $j = 1$, $\nu = 1/2$,

$$G(\hat{g}) = (\hat{g} + i Q_E) [\hat{g} + i (Q_E + Q_i)], \quad (\text{C22})$$

$$V(p, \hat{g}) = -G^{-1/2} \exp(-G^{1/2} p). \quad (\text{C23})$$

Hence, we deduce that

$$\hat{\Delta}_s(\hat{g}) = -\frac{\pi}{(\hat{g} + i Q_E)^{1/2} [\hat{g} + i (Q_E + Q_i)]^{1/2}}, \quad (\text{C24})$$

$$F_s(\hat{g}) = -\frac{2}{[\hat{g} + i (Q_E + Q_e)] (\hat{g} + i Q_E) [\hat{g} + i (Q_E + Q_i)]}. \quad (\text{C25})$$

This response regime is known as the *inertial* regime because the layer response is dominated by ion inertia.^{2,25} The characteristic layer width is $p_* \sim R^{-1}$, which implies that the regime is valid when $P \ll R^3$, $R \gg D$, and $R \gg 1$.

Suppose that $R \ll P p^2$ and $D^2 p^2 \ll 1$. It follows that $k = 2$, $j = 2$, $\nu = 1/4$, and

$$G(\hat{g}) = [\hat{g} + i (Q_E + Q_e)] P_\varphi, \quad (\text{C26})$$

$$V(p, \hat{g}) = -\frac{\Gamma(1/4)}{\sqrt{\pi} 2^{3/4} G^{1/4}} U(0, 2^{1/2} G^{1/4} p), \quad (\text{C27})$$

where $U(a, x) = D_{-a-1/2}(x)$ is a parabolic cylinder function.²⁴ Hence, we deduce that

$$\hat{\Delta}_s(\hat{g}) = -\frac{\pi}{2} \frac{\Gamma(1/4)}{\Gamma(3/4)} \frac{1}{[\hat{g} - i (Q_E + Q_e)]^{1/4} P_\varphi^{1/4}}, \quad (\text{C28})$$

$$F_s(\hat{g}) = -\frac{\pi^{1/2}}{2} \frac{\Gamma(1/4)}{\Gamma(3/4)} \frac{1}{[\hat{g} - i (Q_E + Q_e)]^{3/2} P_\varphi^{1/2}}, \quad (\text{C29})$$

where use has been made of the result

$$\int_0^\infty U(0, x) dx = \frac{\pi}{2^{3/4} \Gamma(3/4)}. \quad (\text{C30})$$

This response regime is known as the *viscous-inertial* regime because the layer response is dominated by ion perpendicular viscosity and ion inertia.² The characteristic layer width is $p_* \sim R^{-1/4} P^{-1/4}$, which implies that the regime is valid when $P \gg R^3$, $P \gg D^4/R$, and $P \gg R^{-3}$.

Suppose, finally, that $R \ll P p^2$ and $D^2 p^2 \gg 1$. It follows that $k = 0$, $j = 1$, $\nu = 1/2$, and

$$G(\hat{g}) = \frac{[\hat{g} + i (Q_E + Q_e)] \iota_e P_\perp}{D^2}, \quad (\text{C31})$$

$$V(p, \hat{g}) = -G^{-1/2} \exp(-G^{1/2} p). \quad (\text{C32})$$

Hence, we deduce that

$$\hat{\Delta}_s = -\frac{\pi D}{[\hat{g} + i(Q_E + Q_e)]^{1/2} \iota_e^{1/2} P_\perp^{1/2}}, \quad (\text{C33})$$

$$F_s(\hat{g}) = -\frac{2 D^2}{[\hat{g} + i(Q_E + Q_e)]^2 \iota_e P_\perp}. \quad (\text{C34})$$

This response regime is known as the *diffusive-inertial* regime because the layer response is dominated by perpendicular energy diffusivity and ion inertia.¹⁶ The characteristic layer width is $p_* \sim R^{-1/2} P^{-1/2} D$, which implies that the regime is valid when $R \ll D$, $P \ll D^4/Q$, and $P \gg D^2/R^2$.

6. Linear Resonant Response Regimes

Figures 1 and 2 show the extents of the various different linear resonant response regimes in R - P space for the cases $D < 1$ and $D > 1$, respectively. Note that Fig. 2 differs slightly from Fig. 2 in Ref. 16 because we are now demanding equal layer thicknesses at the boundaries between the various response regimes.

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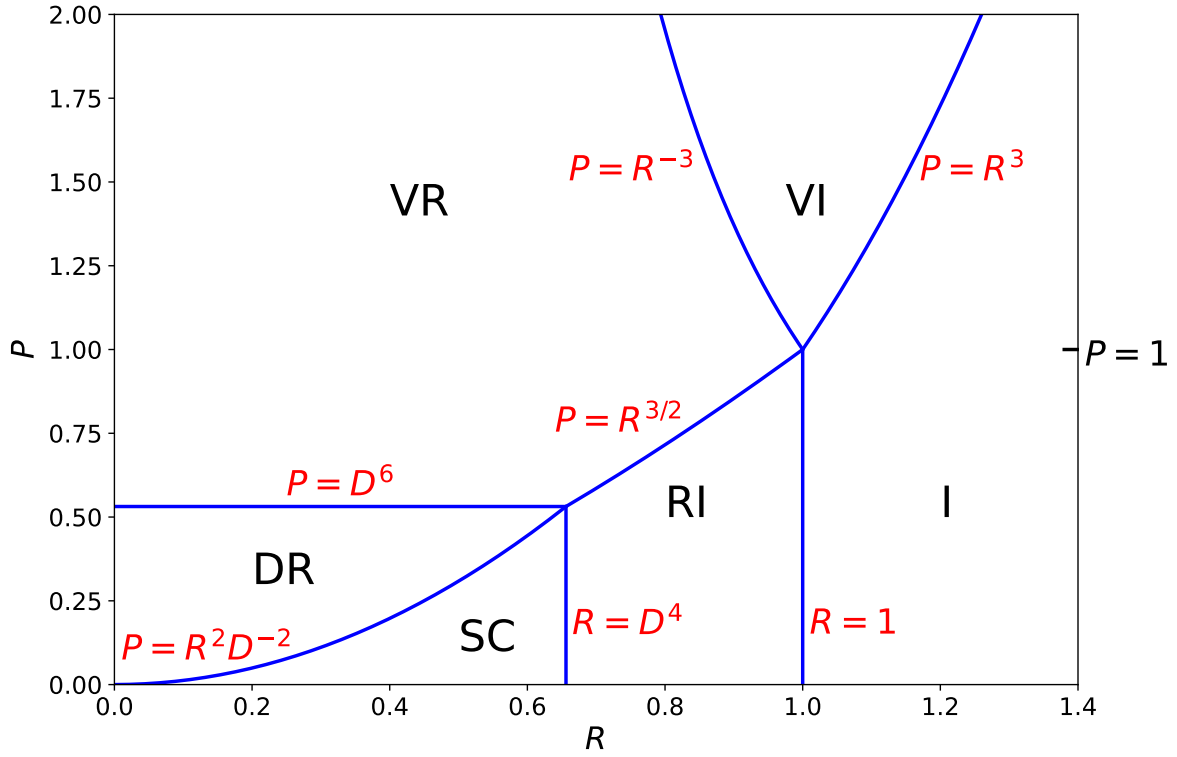


FIG. 1. Linear resonant plasma response regimes in R - P space for the case $D = 0.9$. The various regimes are the diffusive-resistive (DR), the semi-collisional (SC), the resistive-inertial (RI), the viscous-resistive (VR), the viscous-inertial (VI), and the inertial (I).

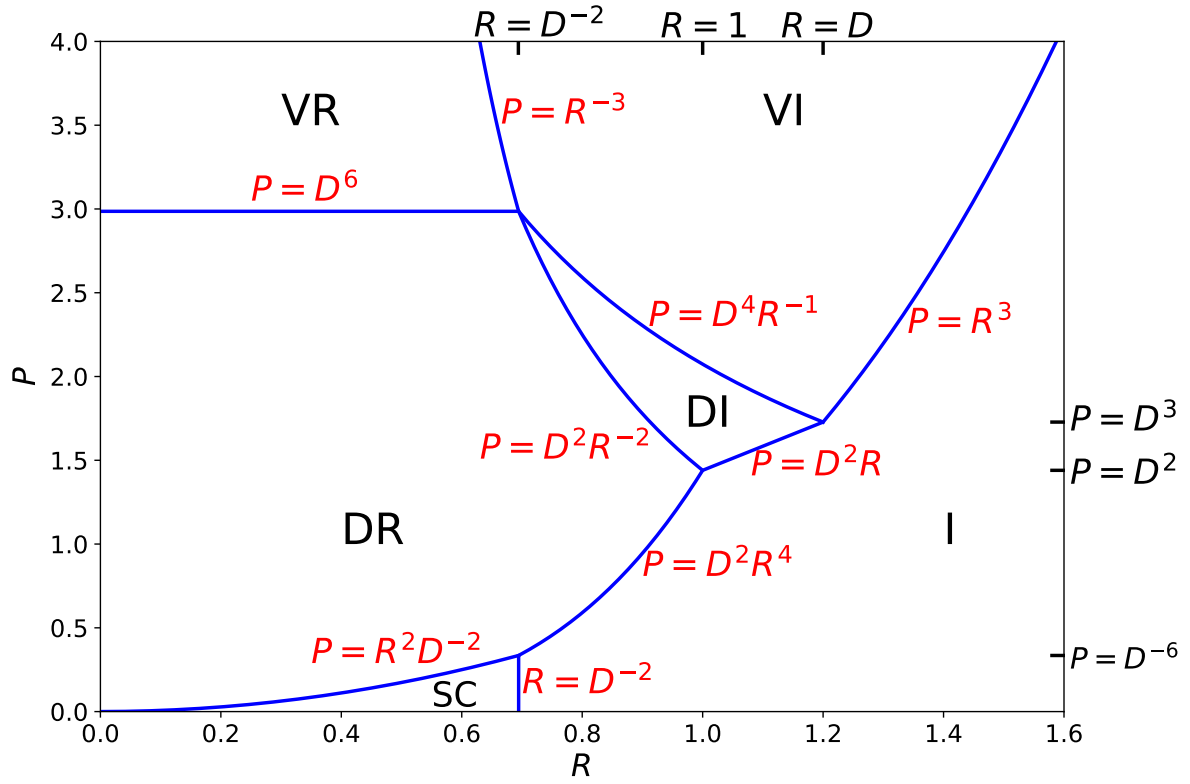


FIG. 2. Linear resonant plasma response regimes in R - P space for the case $D = 1.2$. The various regimes are the diffusive-resistive (DR), the semi-collisional (SC), the diffusive-inertial (DI), the viscous-resistive (VR), the viscous-inertial (VI), and the inertial (I).